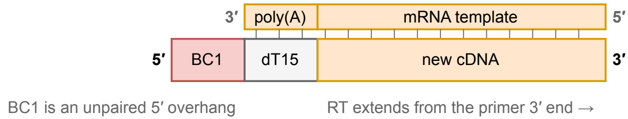
Split-pool barcoding (SPLiT-seq): what happens to the sequence at each step

Yeast-optimized SPLiT-seq (Brettner 2024) The tracked strand runs 5′ to 3′ left to right, so a partner strand runs antiparallel beside it. Steps 1 and 2 are drawn as duplexes because that is where the chemistry acts on two strands.

0. Fixed, permeabilized cell

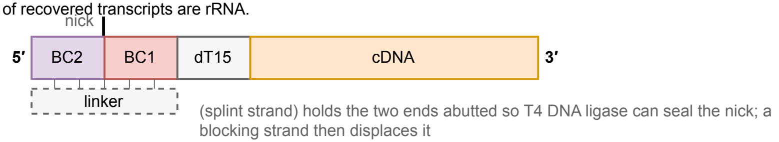
Fix, then digest. 4% formaldehyde (~18 h, 4 °C) freezes the transcriptome and stiffens the cell; only then does zymolyase open the β-glucan wall and Triton X-100 permeabilise the membrane. Digesting first would measure its own stress response.

1. Round 1 barcoded RT, 48 of 96 wells



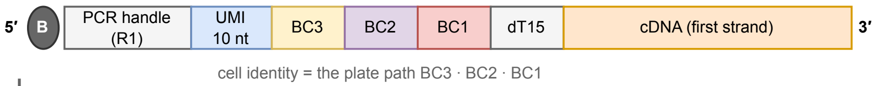
The barcode arrives on the primer, so cDNA is born barcoded. Brettner ran 48 of the 96 wells, ~5,000 cells each; the barcode space below counts the full plate. Every well holds a barcoded 15-dT primer *and* a barcoded random hexamer. Only the dT tract anneals, so BC1 hangs off as a 5′ overhang and RT copies rightward from the primer 3′ end. The hexamer needs no poly(A), so it, not the dT primer, is the arm that transfers to a bacterium – and it is why 93.7% of recovered transcripts are rRNA.

2. Round 2 ligation, 96 wells



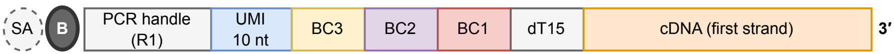
Pool, redistribute, ligate. The linker splints the incoming barcode against the 5′ end of the existing cDNA so T4 ligase can seal the nick (30 min, 37 °C); a blocking strand then retires that well's oligo. Round 3 repeats this chemistry, drawn single-stranded below.

3. Round 3 ligation + UMI + biotin



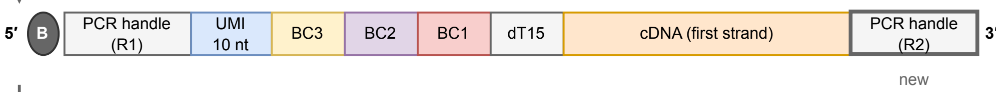
Same chemistry, richer oligo: BC3 plus a 10 nt UMI plus a 5′ biotin, then EDTA terminates. Three rounds of 96 give 96^3 = 884,736 distinct paths; the UMI ($4^{10} \approx 10^6$ states) turns read counts into molecule counts.

4. Sublibraries, lysis streptavidin capture



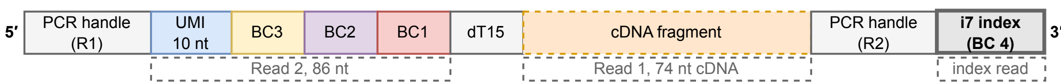
No sequence change – this is a selection. Split into sublibraries of 5,000-20,000 cells, then lyse (SDS plus proteinase K, 55 °C, 2 h); streptavidin beads keep only molecules that received the round-3 oligo, so anything outside the barcoding scheme is never sequenced.

5. Template switch second PCR handle



A template switch on the beads adds a common handle to the far 3′ end, giving the molecule a PCR primer site at both ends. Everything downstream is conventional Illumina library prep.

6. Amplify, fragment, adapter-ligate, index



The Illumina index is barcode round 4 in all but name. The library is amplified from the two handles, fragmented, adapter-ligated and index-PCR'd; the biotin stays behind on the bead. Because cells were split into sublibraries *after* in-cell barcoding, two cells that took identical plate paths are still resolved if they landed in different sublibraries, which is why unique dual indices are mandatory. Three rounds × 24 sublibraries give 21.2 M distinct labels, against 884,736 for the plates alone.